

LINUX LAB ASSIGNMENT

SED · SECURITY LOG ANALYSIS · BASH SCRIPTING

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PART A: AUTHENTICATION LOG

Exercise 1: To Print

Source: *auth.log*

Question	Command / Answer
Print the first log entry.	<code>sed -n '1p' auth.log</code>
Print the fifth log entry.	<code>sed -n '5p' auth.log</code>
Print entries from lines 2 to 6.	<code>sed -n '2,6p' auth.log</code>
Print the last entry.	<code>sed -n '\$p' auth.log</code>
Print every second log entry starting from line 1.	<code>sed -n '1~2p' auth.log</code>
Print all entries containing Failed password.	<code>sed -n '/Failed password/p' auth.log</code>
Print all successful login entries containing Accepted password.	<code>sed -n '/Accepted password/p' auth.log</code>
Print entries from the first failed login to the authentication failure.	<code>sed -n '/Failed password/,/authentication failure/{p; /authentication failure/q}' auth.log</code>

New / Important Symbols Introduced Here

Symbol / Syntax	Meaning
<code>-n</code>	Suppress normal automatic output; print only selected lines.
<code>p</code>	Print the selected/current line.
<code>\$</code>	Represents the last line when used as a line address.

Symbol / Syntax	Meaning
1~2	Start at line 1 and select every 2nd line: 1, 3, 5, ...
/pattern/	Search/select lines containing the pattern.
{ }	Group multiple sed commands.
q	Quit sed immediately; used to stop after the first authentication failure.

```

(gopika@kali)-[~/linux/lab/lab5]
$ nano auth.log

(gopika@kali)-[~/linux/lab/lab5]
$ sed -n '1p' auth.log
Sep 10 09:10:12 server sshd[1201]: Accepted password for admin from 192.168.1.10

(gopika@kali)-[~/linux/lab/lab5]
$ sed -n '5p' auth.log
Sep 10 09:14:33 server sshd[1215]: Invalid user hacker from 172.16.0.8

(gopika@kali)-[~/linux/lab/lab5]
$ sed -n '2,6p' auth.log
Sep 10 09:12:15 server sshd[1205]: Failed password for root from 10.0.0.25
Sep 10 09:12:20 server sshd[1206]: Failed password for root from 10.0.0.25
Sep 10 09:13:01 server sshd[1210]: Accepted password for student from 192.168.1.15
Sep 10 09:14:33 server sshd[1215]: Invalid user hacker from 172.16.0.8
Sep 10 09:15:42 server sshd[1220]: Failed password for admin from 172.16.0.8

(gopika@kali)-[~/linux/lab/lab5]
$ sed -n '$p' auth.log
Sep 10 09:20:10 server sshd[1240]: Connection closed by 10.0.0.25

(gopika@kali)-[~/linux/lab/lab5]
$ sed -n '1~2p' auth.log
Sep 10 09:10:12 server sshd[1201]: Accepted password for admin from 192.168.1.10
Sep 10 09:12:20 server sshd[1206]: Failed password for root from 10.0.0.25
Sep 10 09:14:33 server sshd[1215]: Invalid user hacker from 172.16.0.8
Sep 10 09:16:11 server sudo[1225]: authentication failure for user student
Sep 10 09:18:05 server sshd[1235]: Failed password for root from 10.0.0.30

(gopika@kali)-[~/linux/lab/lab5]
$ sed -n '/Failed password/p' auth.log
Sep 10 09:12:15 server sshd[1205]: Failed password for root from 10.0.0.25
Sep 10 09:12:20 server sshd[1206]: Failed password for root from 10.0.0.25
Sep 10 09:15:42 server sshd[1220]: Failed password for admin from 172.16.0.8
Sep 10 09:18:05 server sshd[1235]: Failed password for root from 10.0.0.30

(gopika@kali)-[~/linux/lab/lab5]
$ sed -n '/Accepted password/p' auth.log
Sep 10 09:10:12 server sshd[1201]: Accepted password for admin from 192.168.1.10
Sep 10 09:13:01 server sshd[1210]: Accepted password for student from 192.168.1.15
Sep 10 09:17:30 server sshd[1230]: Accepted password for analyst from 192.168.1.20

(gopika@kali)-[~/linux/lab/lab5]
$ sed -n '/Failed password/,/authentication failure/p' auth.log
Sep 10 09:12:15 server sshd[1205]: Failed password for root from 10.0.0.25
Sep 10 09:12:20 server sshd[1206]: Failed password for root from 10.0.0.25
Sep 10 09:13:01 server sshd[1210]: Accepted password for student from 192.168.1.15

```

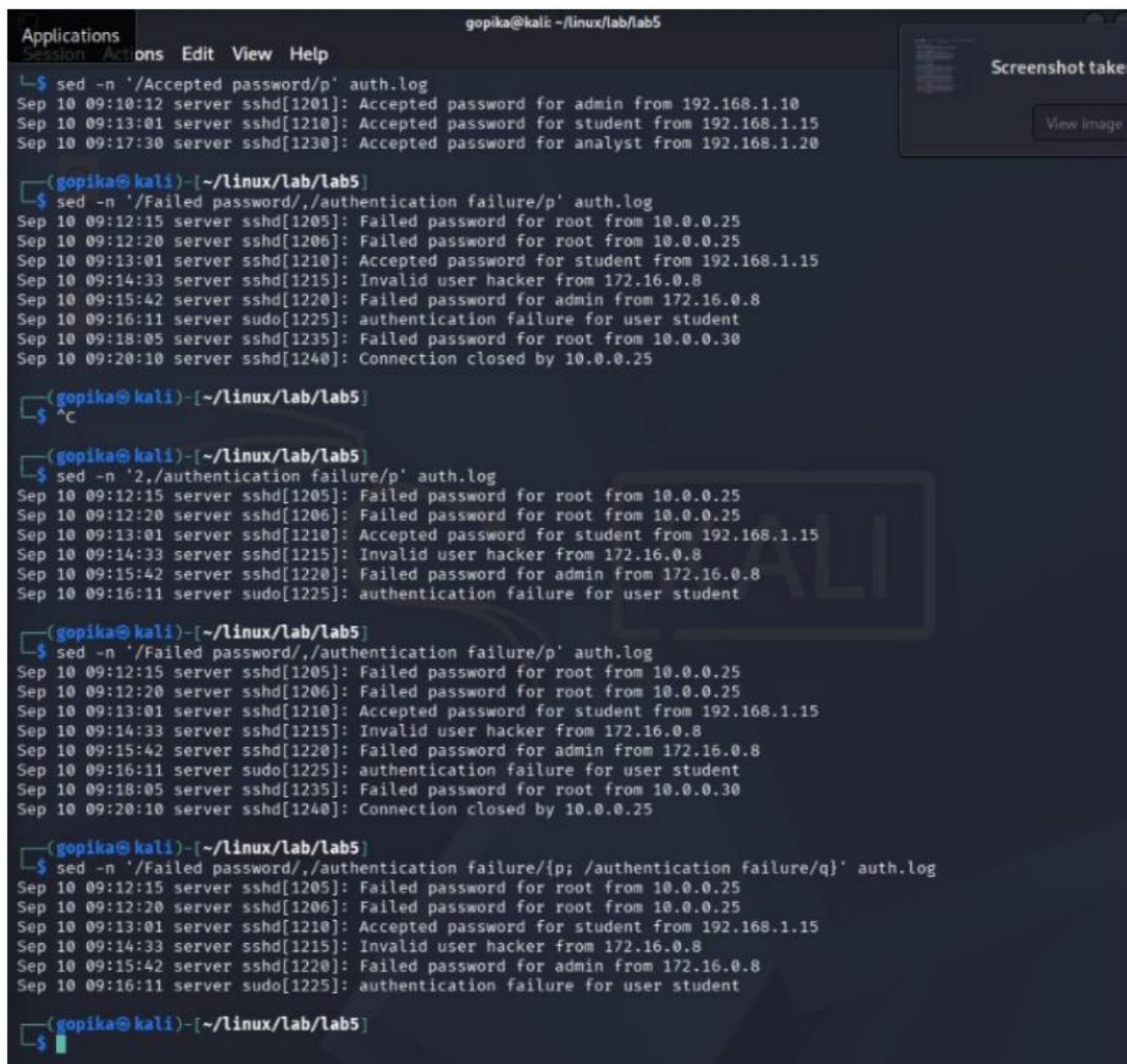
PART A: AUTHENTICATION LOG

Exercise 2: Filtering Security EventsSource: *auth.log*

Question	Command / Answer
Display failed login attempts.	<code>sed -n '/Failed password/p' auth.log</code>
Display successful login attempts.	<code>sed -n '/Accepted password/p' auth.log</code>
Display entries associated with the root account.	<code>sed -n '/root/p' auth.log</code>
Display entries containing Invalid user.	<code>sed -n '/Invalid user/p' auth.log</code>
Display all events related to IP address 172.16.0.8.	<code>sed -n '/172\.16\.0\.8/p' auth.log</code>
Display all events that do not contain Accepted password.	<code>sed -n '/Accepted password/!p' auth.log</code>
Display all events except those generated by sudo.	<code>sed -n '/sudo/!p' auth.log</code>
Display only the entries that contain either Failed password or Invalid user using multiple -e commands.	<code>sed -n -e '/Failed password/p' -e '/Invalid user/p' auth.log</code>

New / Important Symbols Introduced Here

Symbol / Syntax	Meaning
-e	Executes a separate sed expression; allows multiple expressions.
!	Negates the pattern/address; selects non-matching lines.
\.	Matches a literal period, useful in IP addresses.



```

Applications
Session Actions Edit View Help
gopika@kali: ~/linux/lab/lab5

$ sed -n '/Accepted password/p' auth.log
Sep 10 09:10:12 server sshd[1201]: Accepted password for admin from 192.168.1.10
Sep 10 09:13:01 server sshd[1210]: Accepted password for student from 192.168.1.15
Sep 10 09:17:30 server sshd[1230]: Accepted password for analyst from 192.168.1.20

(gopika@kali)~[~/linux/lab/lab5]
$ sed -n '/Failed password/,/authentication failure/p' auth.log
Sep 10 09:12:15 server sshd[1205]: Failed password for root from 10.0.0.25
Sep 10 09:12:20 server sshd[1206]: Failed password for root from 10.0.0.25
Sep 10 09:13:01 server sshd[1210]: Accepted password for student from 192.168.1.15
Sep 10 09:14:33 server sshd[1215]: Invalid user hacker from 172.16.0.8
Sep 10 09:15:42 server sshd[1220]: Failed password for admin from 172.16.0.8
Sep 10 09:16:11 server sudo[1225]: authentication failure for user student
Sep 10 09:18:05 server sshd[1235]: Failed password for root from 10.0.0.30
Sep 10 09:20:10 server sshd[1240]: Connection closed by 10.0.0.25

(gopika@kali)~[~/linux/lab/lab5]
^C

(gopika@kali)~[~/linux/lab/lab5]
$ sed -n '2,/authentication failure/p' auth.log
Sep 10 09:12:15 server sshd[1205]: Failed password for root from 10.0.0.25
Sep 10 09:12:20 server sshd[1206]: Failed password for root from 10.0.0.25
Sep 10 09:13:01 server sshd[1210]: Accepted password for student from 192.168.1.15
Sep 10 09:14:33 server sshd[1215]: Invalid user hacker from 172.16.0.8
Sep 10 09:15:42 server sshd[1220]: Failed password for admin from 172.16.0.8
Sep 10 09:16:11 server sudo[1225]: authentication failure for user student

(gopika@kali)~[~/linux/lab/lab5]
$ sed -n '/Failed password/,/authentication failure/p' auth.log
Sep 10 09:12:15 server sshd[1205]: Failed password for root from 10.0.0.25
Sep 10 09:12:20 server sshd[1206]: Failed password for root from 10.0.0.25
Sep 10 09:13:01 server sshd[1210]: Accepted password for student from 192.168.1.15
Sep 10 09:14:33 server sshd[1215]: Invalid user hacker from 172.16.0.8
Sep 10 09:15:42 server sshd[1220]: Failed password for admin from 172.16.0.8
Sep 10 09:16:11 server sudo[1225]: authentication failure for user student
Sep 10 09:18:05 server sshd[1235]: Failed password for root from 10.0.0.30
Sep 10 09:20:10 server sshd[1240]: Connection closed by 10.0.0.25

(gopika@kali)~[~/linux/lab/lab5]
$ sed -n '/Failed password/,/authentication failure/{p; /authentication failure/q}' auth.log
Sep 10 09:12:15 server sshd[1205]: Failed password for root from 10.0.0.25
Sep 10 09:12:20 server sshd[1206]: Failed password for root from 10.0.0.25
Sep 10 09:13:01 server sshd[1210]: Accepted password for student from 192.168.1.15
Sep 10 09:14:33 server sshd[1215]: Invalid user hacker from 172.16.0.8
Sep 10 09:15:42 server sshd[1220]: Failed password for admin from 172.16.0.8
Sep 10 09:16:11 server sudo[1225]: authentication failure for user student

(gopika@kali)~[~/linux/lab/lab5]
$

```

Ex-2

PART A: AUTHENTICATION LOG

Exercise 3: Removing Irrelevant Entries*Source: auth.log*

Question	Command / Answer
Without modifying the original file, delete successful login entries.	<code>sed '/Accepted password/d' auth.log</code>
Without modifying the original file, delete entries containing Connection closed.	<code>sed '/Connection closed/d' auth.log</code>
Without modifying the original file, delete the first three entries.	<code>sed '1,3d' auth.log</code>
Without modifying the original file, delete entries from lines 4 to 7.	<code>sed '4,7d' auth.log</code>
Without modifying the original file, delete all entries associated with 192.168.1.15.	<code>sed '/192\.168\.1\.15/d' auth.log</code>
Without modifying the original file, delete every second entry starting from the first line.	<code>sed '1~2d' auth.log</code>
Without modifying the original file, delete all events that do not contain Failed password.	<code>sed '/Failed password/!d' auth.log</code>

New / Important Symbols Introduced Here

Symbol / Syntax	Meaning
d	Delete the selected line from the displayed output.
1,3	Selects the range from line 1 through line 3.

```

(gopika@kali)-[~/linux/lab/lab5]
$ sed -n '/Failed password/p' auth.log
Sep 10 09:12:15 server sshd[1205]: Failed password for root from 10.0.0.25
Sep 10 09:12:20 server sshd[1206]: Failed password for root from 10.0.0.25
Sep 10 09:15:42 server sshd[1220]: Failed password for admin from 172.16.0.8
Sep 10 09:18:05 server sshd[1235]: Failed password for root from 10.0.0.30

(gopika@kali)-[~/linux/lab/lab5]
$ sed -n '/Accepted password/p' auth.log
Sep 10 09:10:12 server sshd[1201]: Accepted password for admin from 192.168.1.10
Sep 10 09:13:01 server sshd[1210]: Accepted password for student from 192.168.1.15
Sep 10 09:17:30 server sshd[1230]: Accepted password for analyst from 192.168.1.20

(gopika@kali)-[~/linux/lab/lab5]
$ sed -n '/root/p' auth.log
Sep 10 09:12:15 server sshd[1205]: Failed password for root from 10.0.0.25
Sep 10 09:12:20 server sshd[1206]: Failed password for root from 10.0.0.25
Sep 10 09:18:05 server sshd[1235]: Failed password for root from 10.0.0.30

(gopika@kali)-[~/linux/lab/lab5]
$ sed -n '/Invalid user/p' auth.log
Sep 10 09:14:33 server sshd[1215]: Invalid user hacker from 172.16.0.8

(gopika@kali)-[~/linux/lab/lab5]
$ sed -n '/172.16.0.8/p' auth.log
Sep 10 09:14:33 server sshd[1215]: Invalid user hacker from 172.16.0.8
Sep 10 09:15:42 server sshd[1220]: Failed password for admin from 172.16.0.8

(gopika@kali)-[~/linux/lab/lab5]
$ sed -n '/Accepted password/ip' auth.log
Sep 10 09:12:15 server sshd[1205]: Failed password for root from 10.0.0.25
Sep 10 09:12:20 server sshd[1206]: Failed password for root from 10.0.0.25
Sep 10 09:14:33 server sshd[1215]: Invalid user hacker from 172.16.0.8
Sep 10 09:15:42 server sshd[1220]: Failed password for admin from 172.16.0.8
Sep 10 09:16:11 server sudo[1225]: authentication failure for user student
Sep 10 09:18:05 server sshd[1235]: Failed password for root from 10.0.0.30
Sep 10 09:20:10 server sshd[1240]: Connection closed by 10.0.0.25

(gopika@kali)-[~/linux/lab/lab5]
$ sed -n '/sudo/ip' auth.log
Sep 10 09:10:12 server sshd[1201]: Accepted password for admin from 192.168.1.10
Sep 10 09:12:15 server sshd[1205]: Failed password for root from 10.0.0.25
Sep 10 09:12:20 server sshd[1206]: Failed password for root from 10.0.0.25
Sep 10 09:13:01 server sshd[1210]: Accepted password for student from 192.168.1.15
Sep 10 09:14:33 server sshd[1215]: Invalid user hacker from 172.16.0.8
Sep 10 09:15:42 server sshd[1220]: Failed password for admin from 172.16.0.8
Sep 10 09:17:30 server sshd[1230]: Accepted password for analyst from 192.168.1.20
Sep 10 09:18:05 server sshd[1235]: Failed password for root from 10.0.0.30

(gopika@kali)-[~/linux/lab/lab5]
$ sed -n -e '/Failed password/p' -e '/Invalid user/p' auth.log
Sep 10 09:12:15 server sshd[1205]: Failed password for root from 10.0.0.25
Sep 10 09:12:20 server sshd[1206]: Failed password for root from 10.0.0.25
Sep 10 09:14:33 server sshd[1215]: Invalid user hacker from 172.16.0.8
Sep 10 09:15:42 server sshd[1220]: Failed password for admin from 172.16.0.8
Sep 10 09:18:05 server sshd[1235]: Failed password for root from 10.0.0.30

(gopika@kali)-[~/linux/lab/lab5]

```

Ex-3

PART A: AUTHENTICATION LOG

Exercise 4: Classifying Security Events*Source: auth.log*

Question	Command / Answer
Replace Accepted password with [SUCCESSFUL LOGIN].	<code>sed 's/Accepted password/[SUCCESSFUL LOGIN]/g' auth.log</code>
Replace Failed password with [FAILED LOGIN].	<code>sed 's/Failed password/[FAILED LOGIN]/g' auth.log</code>
Replace Invalid user with [INVALID USER].	<code>sed 's/Invalid user/[INVALID USER]/g' auth.log</code>
Replace authentication failure with [AUTHENTICATION FAILURE].	<code>sed 's/authentication failure/[AUTHENTICATION FAILURE]/g' auth.log</code>
Replace root with PRIVILEGED_USER.	<code>sed 's/root/PRIVILEGED_USER/g' auth.log</code>
Replace every occurrence of server with authentication-server.	<code>sed 's/server/authentication-server/g' auth.log</code>

New / Important Symbols Introduced Here

Symbol / Syntax	Meaning
s	Substitute/replace text.
g	Global replacement; replaces every matching occurrence on a line.

```

(gopika@kali)~[/linux/lab/lab5]
$ sed '4,7d' auth.log
Sep 10 09:10:12 server sshd[1201]: Accepted password for admin from 192.168.1.10
Sep 10 09:12:15 server sshd[1205]: Failed password for root from 10.0.0.25
Sep 10 09:12:20 server sshd[1206]: Failed password for root from 10.0.0.25
Sep 10 09:17:30 server sshd[1230]: Accepted password for analyst from 192.168.1.20
Sep 10 09:18:05 server sshd[1235]: Failed password for root from 10.0.0.30
Sep 10 09:20:10 server sshd[1240]: Connection closed by 10.0.0.25

(gopika@kali)~[/linux/lab/lab5]
$ sed '/192\.\.168\.\.1\.\.15/d' auth.log
Sep 10 09:10:12 server sshd[1201]: Accepted password for admin from 192.168.1.10
Sep 10 09:12:15 server sshd[1205]: Failed password for root from 10.0.0.25
Sep 10 09:12:20 server sshd[1206]: Failed password for root from 10.0.0.25
Sep 10 09:14:33 server sshd[1215]: Invalid user hacker from 172.16.0.8
Sep 10 09:15:42 server sshd[1220]: Failed password for admin from 172.16.0.8
Sep 10 09:16:11 server sudo[1225]: authentication failure for user student
Sep 10 09:17:30 server sshd[1230]: Accepted password for analyst from 192.168.1.20
Sep 10 09:18:05 server sshd[1235]: Failed password for root from 10.0.0.30
Sep 10 09:20:10 server sshd[1240]: Connection closed by 10.0.0.25

(gopika@kali)~[/linux/lab/lab5]
$ sed '1~2d' auth.log
Sep 10 09:12:15 server sshd[1205]: Failed password for root from 10.0.0.25
Sep 10 09:13:01 server sshd[1210]: Accepted password for student from 192.168.1.15
Sep 10 09:15:42 server sshd[1220]: Failed password for admin from 172.16.0.8
Sep 10 09:17:30 server sshd[1230]: Accepted password for analyst from 192.168.1.20
Sep 10 09:20:10 server sshd[1240]: Connection closed by 10.0.0.25

(gopika@kali)~[/linux/lab/lab5]
$ sed '/Failed password/!d' auth.log
Sep 10 09:12:15 server sshd[1205]: Failed password for root from 10.0.0.25
Sep 10 09:12:20 server sshd[1206]: Failed password for root from 10.0.0.25
Sep 10 09:15:42 server sshd[1220]: Failed password for admin from 172.16.0.8
Sep 10 09:18:05 server sshd[1235]: Failed password for root from 10.0.0.30

```

Ex-4

PART A: AUTHENTICATION LOG

Exercise 5: Masking Sensitive Information

Source: *auth.log*

Question	Command / Answer
Replace IP address 10.0.0.25 with X.X.X.X.	<code>sed 's/10\.0\.0\.25/X.X.X.X/g' auth.log</code>
Replace 172.16.0.8 with SUSPICIOUS_IP.	<code>sed 's/172\.16\.0\.8/SUSPICIOUS_IP/g' auth.log</code>
Replace username admin with USER_A.	<code>sed 's/\badmin\b/USER_A/g' auth.log</code>
Replace username student with USER_B.	<code>sed 's/\bstudent\b/USER_B/g' auth.log</code>
Replace all digits with X.	<code>sed 's/[0-9]/X/g' auth.log</code>
Replace periods in IP addresses with hyphens.	<code>sed 's/\./-/g' auth.log</code>
Replace all IP addresses in the sample file with [REDACTED_IP].	<code>sed -E 's/[0-9]+\.[0-9]+\.[0-9]+\.[0-9]+/[REDACTED_IP]/g' auth.log</code>

New / Important Symbols Introduced Here

Symbol / Syntax	Meaning
<code>[0-9]</code>	Character class matching digits 0 through 9.
<code>\b</code>	Word boundary; matches a complete word such as admin or student.
<code>-E</code>	Enables extended regular expressions.
<code>+</code>	One or more occurrences of the preceding pattern.

```

(gopika@kali)-[~/linux/lab/lab5]
$ sed 's/Accepted password/[SUCCESSFUL LOGIN]/g' auth.log
Sep 10 09:10:12 server sshd[1201]: [SUCCESSFUL LOGIN] for admin from 192.168.1.10
Sep 10 09:12:15 server sshd[1205]: Failed password for root from 10.0.0.25
Sep 10 09:12:20 server sshd[1206]: Failed password for root from 10.0.0.25
Sep 10 09:13:01 server sshd[1210]: [SUCCESSFUL LOGIN] for student from 192.168.1.15
Sep 10 09:14:33 server sshd[1215]: Invalid user hacker from 172.16.0.8
Sep 10 09:15:42 server sshd[1220]: Failed password for admin from 172.16.0.8
Sep 10 09:16:11 server sudo[1225]: authentication failure for user student
Sep 10 09:17:30 server sshd[1230]: [SUCCESSFUL LOGIN] for analyst from 192.168.1.20
Sep 10 09:18:05 server sshd[1235]: Failed password for root from 10.0.0.30
Sep 10 09:20:10 server sshd[1240]: Connection closed by 10.0.0.25

(gopika@kali)-[~/linux/lab/lab5]
$ sed 's/Failed password/[FAILED LOGIN]/g' auth.log
Sep 10 09:10:12 server sshd[1201]: Accepted password for admin from 192.168.1.10
Sep 10 09:12:15 server sshd[1205]: [FAILED LOGIN] for root from 10.0.0.25
Sep 10 09:12:20 server sshd[1206]: [FAILED LOGIN] for root from 10.0.0.25
Sep 10 09:13:01 server sshd[1210]: Accepted password for student from 192.168.1.15
Sep 10 09:14:33 server sshd[1215]: Invalid user hacker from 172.16.0.8
Sep 10 09:15:42 server sshd[1220]: [FAILED LOGIN] for admin from 172.16.0.8
Sep 10 09:16:11 server sudo[1225]: authentication failure for user student
Sep 10 09:17:30 server sshd[1230]: Accepted password for analyst from 192.168.1.20
Sep 10 09:18:05 server sshd[1235]: [FAILED LOGIN] for root from 10.0.0.30
Sep 10 09:20:10 server sshd[1240]: Connection closed by 10.0.0.25

(gopika@kali)-[~/linux/lab/lab5]
$ sed -e 's/Accepted password/[SUCCESSFUL LOGIN]/g' \
-e 's/Failed password/[FAILED LOGIN]/g' \
-e 's/Invalid user/[INVALID USER]/g' \
-e 's/authentication failure/[AUTHENTICATION FAILURE]/g' \
-e 's/root/PRIVILEGED_USER/g' \
-e 's/server/authentication-server/g' auth.log
Sep 10 09:10:12 authentication-server sshd[1201]: [SUCCESSFUL LOGIN] for admin from 192.168.1.10
Sep 10 09:12:15 authentication-server sshd[1205]: [FAILED LOGIN] for PRIVILEGED_USER from 10.0.0
Sep 10 09:12:20 authentication-server sshd[1206]: [FAILED LOGIN] for PRIVILEGED_USER from 10.0.0
Sep 10 09:13:01 authentication-server sshd[1210]: [SUCCESSFUL LOGIN] for student from 192.168.1.
Sep 10 09:14:33 authentication-server sshd[1215]: [INVALID USER] hacker from 172.16.0.8
Sep 10 09:15:42 authentication-server sshd[1220]: [FAILED LOGIN] for admin from 172.16.0.8
Sep 10 09:16:11 authentication-server sudo[1225]: [AUTHENTICATION FAILURE] for user student
Sep 10 09:17:30 authentication-server sshd[1230]: [SUCCESSFUL LOGIN] for analyst from 192.168.1.
Sep 10 09:18:05 authentication-server sshd[1235]: [FAILED LOGIN] for PRIVILEGED_USER from 10.0.0
Sep 10 09:20:10 authentication-server sshd[1240]: Connection closed by 10.0.0.25

```

Ex-5

PART B: WEB-SERVER LOG

Exercise 6: Filtering Web Requests

Source: *access.log*

Question	Command / Answer
Print all requests that returned status code 200.	<code>sed -n '/ 200\$/p' access.log</code>
Print all requests that returned status code 401.	<code>sed -n '/ 401\$/p' access.log</code>
Print all requests that returned status code 403.	<code>sed -n '/ 403\$/p' access.log</code>
Print all requests to /admin/login.	<code>sed -n '/\admin\login/p' access.log</code>
Print all POST requests.	<code>sed -n '/POST/p' access.log</code>
Print requests containing ../.	<code>sed -n '/\.\.\./p' access.log</code>
Print requests containing <script>.	<code>sed -n '/<script>/p' access.log</code>
Print requests containing OR.	<code>sed -n '/OR/p' access.log</code>
Print all requests except successful requests.	<code>sed '/ 200\$/d' access.log</code>
Print all requests from 10.0.0.25.	<code>sed -n '/^10\.0\.0\.25 /p' access.log</code>

New / Important Symbols Introduced Here

Symbol / Syntax	Meaning
<code>^</code>	Matches the beginning of a line.

```

(gopika@kali)-[~/linux/lab/lab5]
$ sed 's/[0-9]/X/g' auth.log
Sep XX XX:XX:XX server sshd[XXXX]: Accepted password for admin from XXX.XXX.X.XX
Sep XX XX:XX:XX server sshd[XXXX]: Failed password for root from XX.X.X.XX
Sep XX XX:XX:XX server sshd[XXXX]: Failed password for root from XX.X.X.XX
Sep XX XX:XX:XX server sshd[XXXX]: Accepted password for student from XXX.XXX.X.XX
Sep XX XX:XX:XX server sshd[XXXX]: Invalid user hacker from XXX.XX.X.X
Sep XX XX:XX:XX server sshd[XXXX]: Failed password for admin from XXX.XX.X.X
Sep XX XX:XX:XX server sudo[XXXX]: authentication failure for user student
Sep XX XX:XX:XX server sshd[XXXX]: Accepted password for analyst from XXX.XXX.X.XX
Sep XX XX:XX:XX server sshd[XXXX]: Failed password for root from XX.X.X.XX
Sep XX XX:XX:XX server sshd[XXXX]: Connection closed by XX.X.X.XX

(gopika@kali)-[~/linux/lab/lab5]
$ sed 's/\./-/g' auth.log
Sep 10 09:10:12 server sshd[1201]: Accepted password for admin from 192-168-1-10
Sep 10 09:12:15 server sshd[1205]: Failed password for root from 10-0-0-25
Sep 10 09:12:20 server sshd[1206]: Failed password for root from 10-0-0-25
Sep 10 09:13:01 server sshd[1210]: Accepted password for student from 192-168-1-15
Sep 10 09:14:33 server sshd[1215]: Invalid user hacker from 172-16-0-8
Sep 10 09:15:42 server sshd[1220]: Failed password for admin from 172-16-0-8
Sep 10 09:16:11 server sudo[1225]: authentication failure for user student
Sep 10 09:17:30 server sshd[1230]: Accepted password for analyst from 192-168-1-20
Sep 10 09:18:05 server sshd[1235]: Failed password for root from 10-0-0-30
Sep 10 09:20:10 server sshd[1240]: Connection closed by 10-0-0-25

(gopika@kali)-[~/linux/lab/lab5]
$ sed -E 's/[0-9]+\.[0-9]+\.[0-9]+\.[0-9]+/[REDACTED_IP]/g' auth.log
Sep 10 09:10:12 server sshd[1201]: Accepted password for admin from [REDACTED_IP]
Sep 10 09:12:15 server sshd[1205]: Failed password for root from [REDACTED_IP]
Sep 10 09:12:20 server sshd[1206]: Failed password for root from [REDACTED_IP]
Sep 10 09:13:01 server sshd[1210]: Accepted password for student from [REDACTED_IP]
Sep 10 09:14:33 server sshd[1215]: Invalid user hacker from [REDACTED_IP]
Sep 10 09:15:42 server sshd[1220]: Failed password for admin from [REDACTED_IP]
Sep 10 09:16:11 server sudo[1225]: authentication failure for user student
Sep 10 09:17:30 server sshd[1230]: Accepted password for analyst from [REDACTED_IP]
Sep 10 09:18:05 server sshd[1235]: Failed password for root from [REDACTED_IP]
Sep 10 09:20:10 server sshd[1240]: Connection closed by [REDACTED_IP]

```

Part-b

Ex-6

PART B: WEB-SERVER LOG

Exercise 7: Labelling Suspicious Web Requests

Source: access.log

Question	Command / Answer
Replace status code 200 with [ALLOWED].	sed 's/ 200\$/ [ALLOWED]/' access.log
Replace status code 401 with [UNAUTHORIZED].	sed 's/ 401\$/ [UNAUTHORIZED]/' access.log
Replace status code 403 with [BLOCKED].	sed 's/ 403\$/ [BLOCKED]/' access.log
Replace /admin/login with /RESTRICTED_RESOURCE.	sed 's#/admin/login#/RESTRICTED_RESOURCE#g' access.log
Replace GET with READ_REQUEST.	sed 's/GET/READ_REQUEST/g' access.log
Replace POST with SUBMIT_REQUEST.	sed 's/POST/SUBMIT_REQUEST/g' access.log

New / Important Symbols Introduced Here

Symbol / Syntax	Meaning
#	Alternative substitution delimiter; useful when the text contains / characters.

access.log auth.log

```
(gopika@kali)~[/linux/lab/lab5]
$ cat access.log
192.168.1.10 - GET /index.html 200
10.0.0.25 - GET /admin/login 401
172.16.0.8 - GET /search?id=1'OR'1'='1 403
192.168.1.15 - GET /products.html 200
10.0.0.30 - GET ../../etc/passwd 403
172.16.0.9 - POST /login 401
192.168.1.20 - GET /contact.html 200
10.0.0.25 - POST /admin/login 401
172.16.0.8 - GET /script?input=<script> 403
192.168.1.25 - GET /about.html 200
```

```
(gopika@kali)~[/linux/lab/lab5]
$ sed -n '/ 200$/p' access.log
192.168.1.10 - GET /index.html 200
192.168.1.15 - GET /products.html 200
192.168.1.20 - GET /contact.html 200
192.168.1.25 - GET /about.html 200
```

```
(gopika@kali)~[/linux/lab/lab5]
$ sed -n '/ 401$/p' access.log
10.0.0.25 - GET /admin/login 401
172.16.0.9 - POST /login 401
10.0.0.25 - POST /admin/login 401
```

```
(gopika@kali)~[/linux/lab/lab5]
$ sed -n '/ 403$/p' access.log
172.16.0.8 - GET /search?id=1'OR'1'='1 403
10.0.0.30 - GET ../../etc/passwd 403
172.16.0.8 - GET /script?input=<script> 403
```

```
(gopika@kali)~[/linux/lab/lab5]
$ sed -n '/\admin\/login/p' access.log
10.0.0.25 - GET /admin/login 401
10.0.0.25 - POST /admin/login 401
```

```
(gopika@kali)~[/linux/lab/lab5]
$ sed -n '/POST/p' access.log
172.16.0.9 - POST /login 401
10.0.0.25 - POST /admin/login 401
```

```
(gopika@kali)~[/linux/lab/lab5]
$ sed -n '/\.\.\./p' access.log
10.0.0.30 - GET ../../etc/passwd 403
```

PART B: WEB-SERVER LOG

Exercise 8: Grouping Security Operations

Source: access.log

Question	Command / Answer
For entries containing status code 401: Replace 401 with [UNAUTHORIZED]. Print the modified line.	<pre>sed -n '/ 401\$/{ s/401/[UNAUTHORIZED]/ p }' access.log</pre>
For entries containing status code 403: Replace 403 with [BLOCKED]. Replace the source IP address with [REDACTED_IP]. Print the modified entry.	<pre>sed -n '/ 403\$/{ s/403/[BLOCKED]/ s/^[0-9]\+\.[0-9]\+\.[0-9]\+\.[0-9]\+/[REDACTED_IP]/ p }' access.log</pre>
For requests to /admin/login: Replace the URL with /PROTECTED_PAGE. Replace the source IP with [REDACTED_IP]. Print the result.	<pre>sed -n '/\/admin\/login/{ s#/admin/login#/PROTECTED_PAGE# s/^[0-9]\+\.[0-9]\+\.[0-9]\+\.[0-9]\+/[REDACTED_IP]/ p }' access.log</pre>

All symbols used in this exercise were already introduced in an earlier exercise — see the table there, or the cheat sheet at the end.

```

(gopika@kali)~[/linux/lab/lab5]
$ sed 's#/admin/login#/RESTRICTED_RESOURCE#g' access.log
192.168.1.10 - GET /index.html 200
10.0.0.25 - GET /RESTRICTED_RESOURCE 401
172.16.0.8 - GET /search?id=1'OR'1'='1 403
192.168.1.15 - GET /products.html 200
10.0.0.30 - GET /../../etc/passwd 403
172.16.0.9 - POST /login 401
192.168.1.20 - GET /contact.html 200
10.0.0.25 - POST /RESTRICTED_RESOURCE 401
172.16.0.8 - GET /script?input=<script> 403
192.168.1.25 - GET /about.html 200

(gopika@kali)~[/linux/lab/lab5]
$ sed 's/GET/READ_REQUEST/g' access.log
192.168.1.10 - READ_REQUEST /index.html 200
10.0.0.25 - READ_REQUEST /admin/login 401
172.16.0.8 - READ_REQUEST /search?id=1'OR'1'='1 403
192.168.1.15 - READ_REQUEST /products.html 200
10.0.0.30 - READ_REQUEST /../../etc/passwd 403
172.16.0.9 - POST /login 401
192.168.1.20 - READ_REQUEST /contact.html 200
10.0.0.25 - POST /admin/login 401
172.16.0.8 - READ_REQUEST /script?input=<script> 403
192.168.1.25 - READ_REQUEST /about.html 200

(gopika@kali)~[/linux/lab/lab5]
$ sed 's/GET/READ_REQUEST/g' access.log
192.168.1.10 - READ_REQUEST /index.html 200
10.0.0.25 - READ_REQUEST /admin/login 401
172.16.0.8 - READ_REQUEST /search?id=1'OR'1'='1 403
192.168.1.15 - READ_REQUEST /products.html 200
10.0.0.30 - READ_REQUEST /../../etc/passwd 403
172.16.0.9 - POST /login 401
192.168.1.20 - READ_REQUEST /contact.html 200
10.0.0.25 - POST /admin/login 401
172.16.0.8 - READ_REQUEST /script?input=<script> 403
192.168.1.25 - READ_REQUEST /about.html 200

```

Ex-8

Exercise 9: Adding Incident Labels

Sources: *auth.log* and *access.log*

Question	Command / Answer
Insert the following heading before the first line of <i>auth.log</i> : SECURITY AUTHENTICATION LOG.	<code>sed '1i SECURITY AUTHENTICATION LOG' auth.log</code>
Append the following message after the final entry: END OF SECURITY LOG.	<code>sed '\$a END OF SECURITY LOG' auth.log</code>
Insert [WARNING] before every failed login.	<code>sed '/Failed password/i [WARNING]' auth.log</code>
Append Review this event after every invalid-user entry.	<code>sed '/Invalid user/a Review this event' auth.log</code>
Insert [SUSPICIOUS WEB REQUEST] before every 403 entry.	<code>sed '/ 403\$/i [SUSPICIOUS WEB REQUEST]' access.log</code>
Append [REQUIRES INVESTIGATION] after every 401 entry.	<code>sed '/ 401\$/a [REQUIRES INVESTIGATION]' access.log</code>
Replace all entries containing Connection closed with: [CONNECTION TERMINATED].	<code>sed '/Connection closed/c [CONNECTION TERMINATED]' auth.log</code>
Replace all successful web requests with: [NORMAL WEB REQUEST].	<code>sed '/ 200\$/c [NORMAL WEB REQUEST]' access.log</code>

New / Important Symbols Introduced Here

Symbol / Syntax	Meaning
i	Insert text before the matching line.
a	Append text after the matching line.
c	Change/replace the entire matching line.

```

(gopika@kali)-[~/linux/lab/lab5]
$ sed -n '/ 401$/{
s/401/[UNAUTHORIZED]/
p
}' access.log
10.0.0.25 - GET /admin/login [UNAUTHORIZED]
172.16.0.9 - POST /login [UNAUTHORIZED]
10.0.0.25 - POST /admin/login [UNAUTHORIZED]

(gopika@kali)-[~/linux/lab/lab5]
$ sed -n '/ 403$/{
s/403/[BLOCKED]/
s/^[0-9]\+\.[0-9]\+\.[0-9]\+\.[0-9]\+/[REDACTED_IP]/
p
}' access.log
[REDACTED_IP] - GET /search?id=1'OR'1'='1 [BLOCKED]
[REDACTED_IP] - GET ../../etc/passwd [BLOCKED]
[REDACTED_IP] - GET /script?input=<script> [BLOCKED]

(gopika@kali)-[~/linux/lab/lab5]
$ sed -n '/\admin\login/{
s#/admin/login#/PROTECTED_PAGE#
s/^[0-9]\+\.[0-9]\+\.[0-9]\+\.[0-9]\+/[REDACTED_IP]/
p
}' access.log
[REDACTED_IP] - GET /PROTECTED_PAGE 401
[REDACTED_IP] - POST /PROTECTED_PAGE 401

```

Ex-9

Exercise 10: Separating Log Entries with G

Sources: *auth.log* and *access.log*

Question	Command / Answer
Add a blank line after every authentication event.	<code>sed 'G' auth.log</code>
Add a blank line after every failed login.	<code>sed '/Failed password/G' auth.log</code>
Add a blank line after every blocked web request.	<code>sed '/ 403\$/G' access.log</code>
Add a blank line after every second entry starting with the first entry.	<code>sed '1~2G' auth.log</code>

New / Important Symbols Introduced Here

Symbol / Syntax	Meaning
G	Appends the hold space to the pattern space; with an empty hold space it is used here to add a blank line.

```
(gopika@kali)~[~/linux/lab/lab5]
$ sed '/Connection closed/c [CONNECTION TERMINATED]' auth.log
Sep 10 09:10:12 server sshd[1201]: Accepted password for admin from 192.168.1.10
Sep 10 09:12:15 server sshd[1205]: Failed password for root from 10.0.0.25
Sep 10 09:12:20 server sshd[1206]: Failed password for root from 10.0.0.25
Sep 10 09:13:01 server sshd[1210]: Accepted password for student from 192.168.1.15
Sep 10 09:14:33 server sshd[1215]: Invalid user hacker from 172.16.0.8
Sep 10 09:15:42 server sshd[1220]: Failed password for admin from 172.16.0.8
Sep 10 09:16:11 server sudo[1225]: authentication failure for user student
Sep 10 09:17:30 server sshd[1230]: Accepted password for analyst from 192.168.1.20
Sep 10 09:18:05 server sshd[1235]: Failed password for root from 10.0.0.30
[CONNECTION TERMINATED]

(gopika@kali)~[~/linux/lab/lab5]
$ sed '/ 200$/c [NORMAL WEB REQUEST]' access.log
[NORMAL WEB REQUEST]
10.0.0.25 - GET /admin/login 401
172.16.0.8 - GET /search?id=1'OR'1'='1 403
[NORMAL WEB REQUEST]
10.0.0.30 - GET /../..etc/passwd 403
172.16.0.9 - POST /login 401
[NORMAL WEB REQUEST]
10.0.0.25 - POST /admin/login 401
172.16.0.8 - GET /script?input=<script> 403
[NORMAL WEB REQUEST]
```

Ex-10

Exercise 11: Writing Selected Log Entries to Files

Sources: *auth.log* and *access.log*

Question	Command / Answer
Write failed logins to failed_logins.txt.	<code>sed -n '/Failed password/w failed_logins.txt' auth.log</code>
Write successful logins to successful_logins.txt.	<code>sed -n '/Accepted password/w successful_logins.txt' auth.log</code>
Write invalid-user events to invalid_users.txt.	<code>sed -n '/Invalid user/w invalid_users.txt' auth.log</code>
Write 401 web events to unauthorized_requests.txt.	<code>sed -n '/ 401\$/w unauthorized_requests.txt' access.log</code>
Write 403 web events to blocked_requests.txt.	<code>sed -n '/ 403\$/w blocked_requests.txt' access.log</code>
Write all events related to 172.16.0.8 to suspicious_ip.txt.	<code>sed -n '/172\.16\.0\.8/w suspicious_ip.txt' auth.log</code>

New / Important Symbols Introduced Here

Symbol / Syntax	Meaning
w	Writes selected lines to the specified file.

```

(gopika@kali)-[~/linux/lab/lab5]
$ sed 'G' auth.log
Sep 10 09:10:12 server sshd[1201]: Accepted password for admin from 192.168.1.10
Sep 10 09:12:15 server sshd[1205]: Failed password for root from 10.0.0.25
Sep 10 09:12:20 server sshd[1206]: Failed password for root from 10.0.0.25
Sep 10 09:13:01 server sshd[1210]: Accepted password for student from 192.168.1.15
Sep 10 09:14:33 server sshd[1215]: Invalid user hacker from 172.16.0.8
Sep 10 09:15:42 server sshd[1220]: Failed password for admin from 172.16.0.8
Sep 10 09:16:11 server sudo[1225]: authentication failure for user student
Sep 10 09:17:30 server sshd[1230]: Accepted password for analyst from 192.168.1.20
Sep 10 09:18:05 server sshd[1235]: Failed password for root from 10.0.0.30
Sep 10 09:20:10 server sshd[1240]: Connection closed by 10.0.0.25

(gopika@kali)-[~/linux/lab/lab5]
$ sed '/Failed password/G' auth.log
Sep 10 09:10:12 server sshd[1201]: Accepted password for admin from 192.168.1.10
Sep 10 09:12:15 server sshd[1205]: Failed password for root from 10.0.0.25
Sep 10 09:12:20 server sshd[1206]: Failed password for root from 10.0.0.25
Sep 10 09:13:01 server sshd[1210]: Accepted password for student from 192.168.1.15
Sep 10 09:14:33 server sshd[1215]: Invalid user hacker from 172.16.0.8
Sep 10 09:15:42 server sshd[1220]: Failed password for admin from 172.16.0.8
Sep 10 09:16:11 server sudo[1225]: authentication failure for user student
Sep 10 09:17:30 server sshd[1230]: Accepted password for analyst from 192.168.1.20
Sep 10 09:18:05 server sshd[1235]: Failed password for root from 10.0.0.30
Sep 10 09:20:10 server sshd[1240]: Connection closed by 10.0.0.25

(gopika@kali)-[~/linux/lab/lab5]
$ sed '/ 403$/G' access.log
192.168.1.10 - GET /index.html 200
10.0.0.25 - GET /admin/login 401
172.16.0.8 - GET /search?id=1'OR'1'='1 403
192.168.1.15 - GET /products.html 200

(gopika@kali)-[~/linux/lab/lab5]
$ sed '1~2G' auth.log
Sep 10 09:10:12 server sshd[1201]: Accepted password for admin from 192.168.1.10
Sep 10 09:12:15 server sshd[1205]: Failed password for root from 10.0.0.25
Sep 10 09:12:20 server sshd[1206]: Failed password for root from 10.0.0.25
Sep 10 09:13:01 server sshd[1210]: Accepted password for student from 192.168.1.15
Sep 10 09:14:33 server sshd[1215]: Invalid user hacker from 172.16.0.8
Sep 10 09:15:42 server sshd[1220]: Failed password for admin from 172.16.0.8
Sep 10 09:16:11 server sudo[1225]: authentication failure for user student
Sep 10 09:17:30 server sshd[1230]: Accepted password for analyst from 192.168.1.20
Sep 10 09:18:05 server sshd[1235]: Failed password for root from 10.0.0.30
Sep 10 09:20:10 server sshd[1240]: Connection closed by 10.0.0.25

```

Ex-11

Exercise 12: Combined Security Operations

Sources: auth.log and access.log

Question	Command / Answer
Label successful and failed logins.	<code>sed -e 's/Accepted password/[SUCCESSFUL LOGIN]/g' -e 's/Failed password/[FAILED LOGIN]/g' auth.log</code>
Label 401 and 403 web requests.	<code>sed -e 's/ 401\$/ [UNAUTHORIZED]/' -e 's/ 403\$/ [BLOCKED]/' access.log</code>
Delete successful logins and print failed logins.	<code>sed -n '/Failed password/p' auth.log</code>
Replace root with [PRIVILEGED_USER] and mask all IP addresses.	<code>sed -E -e 's/root/[PRIVILEGED_USER]/g' -e 's/[0-9]+\.[0-9]+\.[0-9]+\.[0-9]+/[REDACTED_IP]/g' auth.log</code>
Replace GET and POST with READ_REQUEST and SUBMIT_REQUEST.	<code>sed -e 's/GET/READ_REQUEST/g' -e 's/POST/SUBMIT_REQUEST/g' access.log</code>

All symbols used in this exercise were already introduced in an earlier exercise — see the table there, or the cheat sheet at the end.

```
(gopika@kali)-[~/linux/lab/lab5]
$ sed -n '/Failed password/w failed_logins.txt' auth.log

(gopika@kali)-[~/linux/lab/lab5]
$ sed -n '/Accepted password/w successful_logins.txt' auth.log

(gopika@kali)-[~/linux/lab/lab5]
$ sed -n '/Invalid user/w invalid_users.txt' auth.log

(gopika@kali)-[~/linux/lab/lab5]
$ sed -n '/ 401$/w unauthorized_requests.txt' access.log

(gopika@kali)-[~/linux/lab/lab5]
$ sed -n '/ 403$/w blocked_requests.txt' access.log

(gopika@kali)-[~/linux/lab/lab5]
$ sed -n '/172\.\16\.\0\.\8/w suspicious_ip.txt' auth.log

(gopika@kali)-[~/linux/lab/lab5]
$ ls
access.log  blocked_requests.txt  invalid_users.txt  suspicious_ip.txt
auth.log    failed_logins.txt      successful_logins.txt  unauthorized_requests.txt
```

Ex-12

Exercise 13: Create and Execute security_report.sed

Source: auth.log

Question	Command / Answer
Create security_report.sed containing commands to insert SECURITY INCIDENT REPORT before the first line.	1i SECURITY INCIDENT REPORT
Replace Failed password with [FAILED LOGIN].	s/Failed password/[FAILED LOGIN]/g
Replace Invalid user with [INVALID USER ATTEMPT].	s/Invalid user/[INVALID USER ATTEMPT]/g
Replace Accepted password with [SUCCESSFUL LOGIN].	s/Accepted password/[SUCCESSFUL LOGIN]/g
Replace all sample IP addresses with [REDACTED_IP].	s/[0-9]\{1,3\}\.[0-9]\{1,3\}\.[0-9]\{1,3\}\.[0-9]\{1,3\}/[REDACTED_IP]/g
Append END OF INCIDENT REPORT after the last line.	\$a END OF INCIDENT REPORT
Execute it using: sed -f security_report.sed auth.log	sed -f security_report.sed auth.log
Save the output: sed -f security_report.sed auth.log > incident_report.txt	sed -f security_report.sed auth.log > incident_report.txt

New / Important Symbols Introduced Here

Symbol / Syntax	Meaning
-f	Reads and executes sed commands from a command file.
>	Redirects output to a file, creating or overwriting it.

```

(gopika@kali)-[~/linux/lab/lab5]
$ sed -e 's/Accepted password/[SUCCESSFUL LOGIN]/g' \
-e 's/Failed password/[FAILED LOGIN]/g' auth.log
Sep 10 09:10:12 server sshd[1201]: [SUCCESSFUL LOGIN] for admin from 192.168.1.10
Sep 10 09:12:15 server sshd[1205]: [FAILED LOGIN] for root from 10.0.0.25
Sep 10 09:12:20 server sshd[1206]: [FAILED LOGIN] for root from 10.0.0.25
Sep 10 09:13:01 server sshd[1210]: [SUCCESSFUL LOGIN] for student from 192.168.1.15
Sep 10 09:14:33 server sshd[1215]: Invalid user hacker from 172.16.0.8
Sep 10 09:15:42 server sshd[1220]: [FAILED LOGIN] for admin from 172.16.0.8
Sep 10 09:16:11 server sudo[1225]: authentication failure for user student
Sep 10 09:17:30 server sshd[1230]: [SUCCESSFUL LOGIN] for analyst from 192.168.1.20
Sep 10 09:18:05 server sshd[1235]: [FAILED LOGIN] for root from 10.0.0.30
Sep 10 09:20:10 server sshd[1240]: Connection closed by 10.0.0.25

(gopika@kali)-[~/linux/lab/lab5]
$ sed -e 's/ 401$/ [UNAUTHORIZED]/' \
-e 's/ 403$/ [BLOCKED]/' access.log
192.168.1.10 - GET /index.html 200
10.0.0.25 - GET /admin/login [UNAUTHORIZED]
172.16.0.8 - GET /search?id=1'OR'1='1 [BLOCKED]
192.168.1.15 - GET /products.html 200
10.0.0.30 - GET ../../etc/passwd [BLOCKED]
172.16.0.9 - POST /login [UNAUTHORIZED]
192.168.1.20 - GET /contact.html 200
10.0.0.25 - POST /admin/login [UNAUTHORIZED]
172.16.0.8 - GET /script?input=<script> [BLOCKED]
192.168.1.25 - GET /about.html 200

(gopika@kali)-[~/linux/lab/lab5]
$ sed 's/Accepted password/d; /Failed password/id' auth.log
Sep 10 09:12:15 server sshd[1205]: Failed password for root from 10.0.0.25
Sep 10 09:12:20 server sshd[1206]: Failed password for root from 10.0.0.25
Sep 10 09:15:42 server sshd[1220]: Failed password for admin from 172.16.0.8
Sep 10 09:18:05 server sshd[1235]: Failed password for root from 10.0.0.30

(gopika@kali)-[~/linux/lab/lab5]
$ sed -E \
-e 's/root/[PRIVILEGED_USER]/g' \
-e 's/[0-9]+\.[0-9]+\.[0-9]+\.[0-9]+/[REDACTED_IP]/g' \
auth.log
Sep 10 09:10:12 server sshd[1201]: Accepted password for admin from [REDACTED_IP]
Sep 10 09:12:15 server sshd[1205]: Failed password for [PRIVILEGED_USER] from [REDACTED_IP]
Sep 10 09:12:20 server sshd[1206]: Failed password for [PRIVILEGED_USER] from [REDACTED_IP]
Sep 10 09:13:01 server sshd[1210]: Accepted password for student from [REDACTED_IP]
Sep 10 09:14:33 server sshd[1215]: Invalid user hacker from [REDACTED_IP]
Sep 10 09:15:42 server sshd[1220]: Failed password for admin from [REDACTED_IP]
Sep 10 09:16:11 server sudo[1225]: authentication failure for user student

(gopika@kali)-[~/linux/lab/lab5]
$ sed -e 's/GET/READ_REQUEST/g' \
-e 's/POST/SUBMIT_REQUEST/g' access.log
192.168.1.10 - READ_REQUEST /index.html 200
10.0.0.25 - READ_REQUEST /admin/login 401
172.16.0.8 - READ_REQUEST /search?id=1'OR'1='1 403
192.168.1.15 - READ_REQUEST /products.html 200
10.0.0.30 - READ_REQUEST ../../etc/passwd 403
172.16.0.9 - SUBMIT_REQUEST /login 401
192.168.1.20 - READ_REQUEST /contact.html 200
10.0.0.25 - SUBMIT_REQUEST /admin/login 401
172.16.0.8 - READ_REQUEST /script?input=<script> 403
192.168.1.25 - READ_REQUEST /about.html 200

```

Ex 13

Exercise 14: Modifying a Security Configuration File

Source: *security.conf*

Question	Command / Answer
Create security.conf with: PermitRootLogin yes; PasswordAuthentication yes; MaxAuthTries 6; FirewallStatus disabled; LogLevel INFO; GuestLogin enabled.	<code>nano security.conf</code>
First make a backup.	<code>cp security.conf security.conf.backup</code>
Change PermitRootLogin yes to PermitRootLogin no.	<code>sed -i 's/PermitRootLogin yes/PermitRootLogin no/' security.conf</code>
Change MaxAuthTries 6 to MaxAuthTries 3.	<code>sed -i 's/MaxAuthTries 6/MaxAuthTries 3/' security.conf</code>
Change FirewallStatus disabled to FirewallStatus enabled.	<code>sed -i 's/FirewallStatus disabled/FirewallStatus enabled/' security.conf</code>
Change LogLevel INFO to LogLevel VERBOSE.	<code>sed -i 's/LogLevel INFO/LogLevel VERBOSE/' security.conf</code>
Change GuestLogin enabled to GuestLogin disabled.	<code>sed -i 's/GuestLogin enabled/GuestLogin disabled/' security.conf</code>
Use the safer backup form: sed -i.bak 'sed-command' security.conf	<code>sed -i.bak 's/PermitRootLogin yes/PermitRootLogin no/' security.conf</code>
Check the final configuration.	<code>cat security.conf</code>

New / Important Symbols Introduced Here

Symbol / Syntax	Meaning
-i	Edit the file in place.
.bak	Creates a backup when used with sed -i.bak.
cp	Copies a file; used here to create a manual backup.

```

gopika@kali: ~/linux/lab/lab5
Session Actions Edit View Help
GNU nano 9.0 security_report.sed
11 SECURITY INCIDENT REPORT
s/Failed password/[FAILED LOGIN]/g
s/Invalid user/[INVALID USER ATTEMPT]/g
s/Accepted password/[SUCCESSFUL LOGIN]/g
s/[0-9]\{1,3\}\.[0-9]\{1,3\}\.[0-9]\{1,3\}\.[0-9]\{1,3\}/[REDACTED_IP]/g
$a END OF INCIDENT REPORT

(gopika@kali)~[~/linux/lab/lab5]
$ nano security_report.sed

(gopika@kali)~[~/linux/lab/lab5]
$ ls
access.log  blocked_requests.txt  invalid_users.txt  successful_logins.txt  unauthorized_requests.txt
auth.log    failed_logins.txt      security_report.sed  suspicious_ip.txt

(gopika@kali)~[~/linux/lab/lab5]
$ sed -f security_report.sed auth.log
SECURITY INCIDENT REPORT
Sep 10 09:10:12 server sshd[1201]: [SUCCESSFUL LOGIN] for admin from [REDACTED_IP]
Sep 10 09:12:15 server sshd[1205]: [FAILED LOGIN] for root from [REDACTED_IP]
Sep 10 09:12:20 server sshd[1206]: [FAILED LOGIN] for root from [REDACTED_IP]
Sep 10 09:13:01 server sshd[1210]: [SUCCESSFUL LOGIN] for student from [REDACTED_IP]
Sep 10 09:14:33 server sshd[1215]: [INVALID USER ATTEMPT] hacker from [REDACTED_IP]
Sep 10 09:15:42 server sshd[1220]: [FAILED LOGIN] for admin from [REDACTED_IP]
Sep 10 09:16:11 server sudo[1225]: authentication failure for user student
Sep 10 09:17:30 server sshd[1230]: [SUCCESSFUL LOGIN] for analyst from [REDACTED_IP]
Sep 10 09:18:05 server sshd[1235]: [FAILED LOGIN] for root from [REDACTED_IP]
Sep 10 09:20:10 server sshd[1240]: Connection closed by [REDACTED_IP]
END OF INCIDENT REPORT

(gopika@kali)~[~/linux/lab/lab5]
$ sed -f security_report.sed auth.log > incident_report.txt

(gopika@kali)~[~/linux/lab/lab5]
$ cat incident_report.txt
SECURITY INCIDENT REPORT
Sep 10 09:10:12 server sshd[1201]: [SUCCESSFUL LOGIN] for admin from [REDACTED_IP]
Sep 10 09:12:15 server sshd[1205]: [FAILED LOGIN] for root from [REDACTED_IP]
Sep 10 09:12:20 server sshd[1206]: [FAILED LOGIN] for root from [REDACTED_IP]
Sep 10 09:13:01 server sshd[1210]: [SUCCESSFUL LOGIN] for student from [REDACTED_IP]
Sep 10 09:14:33 server sshd[1215]: [INVALID USER ATTEMPT] hacker from [REDACTED_IP]
Sep 10 09:15:42 server sshd[1220]: [FAILED LOGIN] for admin from [REDACTED_IP]
Sep 10 09:16:11 server sudo[1225]: authentication failure for user student
Sep 10 09:17:30 server sshd[1230]: [SUCCESSFUL LOGIN] for analyst from [REDACTED_IP]
Sep 10 09:18:05 server sshd[1235]: [FAILED LOGIN] for root from [REDACTED_IP]
Sep 10 09:20:10 server sshd[1240]: Connection closed by [REDACTED_IP]
END OF INCIDENT REPORT

```

Ex-14

EXERCISE 15: BASH SCRIPT

security_log_analyser.sh

Script: security_log_analyser.sh

Question	Command / Answer
Accept a log filename as a command-line argument.	<code>LOGFILE="\$1"</code>
Check whether the argument is provided.	<code>if [\$# -eq 0]; then echo "Usage: \$0 <logfile>"; exit 1; fi</code>
Check whether the file exists.	<code>if [! -f "\$LOGFILE"]; then echo "Error: File does not exist."; exit 1; fi</code>
Extract failed login attempts.	<code>sed -n '/Failed password/p' "\$LOGFILE"</code>
Extract invalid-user attempts.	<code>sed -n '/Invalid user/p' "\$LOGFILE"</code>
Replace sensitive IP addresses with [REDACTED_IP].	<code>sed -E 's/[0-9]+\.[0-9]+\.[0-9]+\.[0-9]+/[REDACTED_IP]/g'</code>
Add the heading SECURITY ALERT REPORT.	<code>echo "SECURITY ALERT REPORT" > "\$REPORT"</code>
Add [ALERT] before suspicious entries.	<code>sed 's/^/[ALERT] /'</code>
Append END OF REPORT.	<code>echo "END OF REPORT" >> "\$REPORT"</code>
Save the result in security_alert_report.txt and display the generated report.	<code>REPORT="security_alert_report.txt"</code> <code>cat "\$REPORT"</code>

New / Important Symbols Introduced Here

Symbol / Syntax	Meaning
\$1	First command-line argument; here it is the log filename.
\$#	Number of command-line arguments supplied.
if	Starts a conditional statement.
-eq	Tests numeric equality.
>>	Appends output to a file.

No screenshot was included for this exercise in the source material — run the script yourself and paste your terminal screenshot here before submission.

Quick reference — SED & Bash commands used across this assignment

Command / Symbol	Quick Meaning
<code>sed -n '1p' file</code>	Print line 1
<code>sed -n '2,6p' file</code>	Print lines 2–6
<code>sed -n '\$p' file</code>	Print last line
<code>sed -n '1~2p' file</code>	Print lines 1, 3, 5, ...
<code>sed -n '/pattern/p' file</code>	Print matching lines
<code>sed -n '/pattern/!p' file</code>	Print non-matching lines
<code>sed '/pattern/d' file</code>	Delete matching lines from output
<code>sed 's/old/new/g' file</code>	Replace text globally
<code>sed 's#old#new#g' file</code>	Substitute using # delimiter
<code>sed '/pattern/i text' file</code>	Insert before matching line
<code>sed '/pattern/a text' file</code>	Append after matching line
<code>sed '/pattern/c text' file</code>	Replace entire matching line
<code>sed '/pattern/G' file</code>	Add blank line after matching line
<code>sed -n '/pattern/w out.txt' file</code>	Write matching lines to a file
<code>sed -f script.sed file</code>	Run commands stored in a sed file
<code>sed -i 's/old/new/' file</code>	Edit file in place
<code>sed -i.bak 's/old/new/' file</code>	Edit and create .bak backup
<code>sed -E 'regex' file</code>	Use extended regular expressions
<code>\$1 / \$#</code>	First argument / number of arguments
<code>> / >></code>	Overwrite/create / append to file
<code> </code>	Pass output to the next command